

Clustering similar books

Once book descriptions are vectorized, we can group similar books together. This is done using clustering algorithms.

Clustering creates thematic groups such as:

- Productivity and leadership books
- Epic fantasy novels
- Crime thrillers
- Technology and programming

This improves recommendations by providing contextual neighbourhoods.

KMeans clustering: I used KMeans, one of the most popular unsupervised clustering methods. The steps to follow are:

- Choose number of clusters (k).
 - Assign books to clusters.
 - Books in the same cluster share similarity.
- To select the optimal k, I applied the elbow method.

Dimensionality reduction for visualisation: Since TF-IDF vectors are high-dimensional, I used PCA (Principal Component Analysis). This allowed me to visualise clusters in a 2D space. Clustering provides structure to the recommendation space.

Building recommendation models

I implemented three recommendation approaches.

Content-based filtering: This uses cosine similarity between TF-IDF vectors.

The workflow is:

- User selects a book.
- Similarity with all other books is computed.
- The top N most similar books are recommended.

This method leads to highly relevant recommendations. However, the limitation is that these can become too narrow (only very similar items).

Cluster-based recommendation: In this method, books are recommended from within the same cluster.

The workflow is:

- Identify the cluster of the input book.
- Recommend popular/highly rated books in that cluster.

This method is fast and gives diverse recommendations within genre themes. A limitation is that cluster boundaries may be imperfect.

Hybrid recommendation model: The hybrid model combines both content similarity (precision) and cluster context (diversity).

Hybrid logic is:

- Top recommendations from cosine similarity.
- Additional recommendations from same cluster.

This balances relevance with discovery.

Hybrid models are used widely because they reduce the monotony in suggestions.

Evaluation of recommendation performance

To evaluate recommendation quality, I used metrics such as *Precision@5*, while *Recall@5* was used to find out how many of the top 5 recommendations are truly relevant.

The Content model gave more precise recommendations, the Cluster model gave broader recall and the Hybrid model gave the best balance.

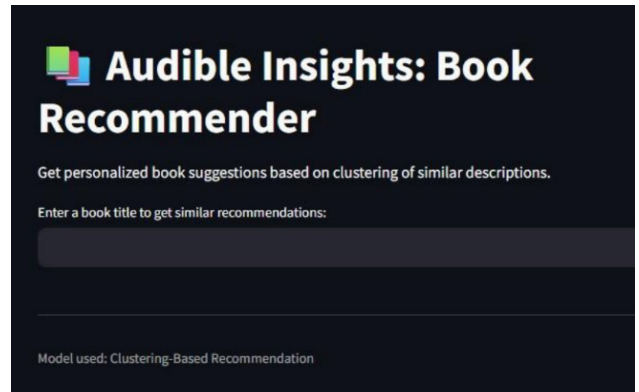


Figure 1: Audible Insights: Book recommendations

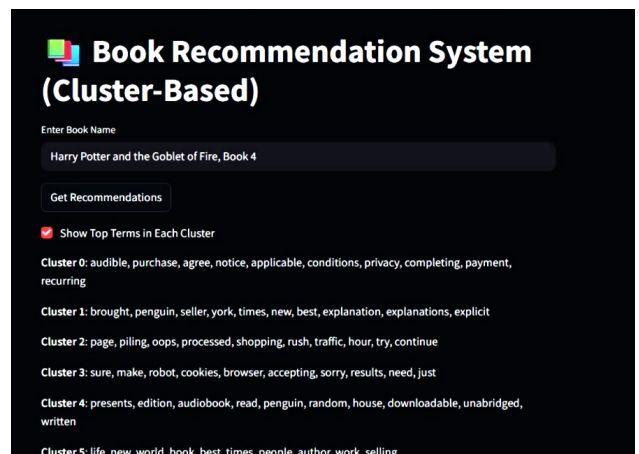


Figure 2: Searching for recommendations



Figure 3: Recommended books